Engineering - Undergraduate Level Assessment 9 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. In decimal number system what is MSD
   (a) First two digits from right to left
   (b) Median of all digits
   (c) First digit from right to left
   (d) First digit from left to right
   (e) Mode of all digits

2. Two long parallel conductors carry 100 A. If the conductors are separated by 20 mm, the force per meter of length of each conductor will be
   (a) 100 N.
   (b) 0.01 N.
   (c) 10 N.
   (d) 0.5 N.
   (e) 0.001 N.

3. The bandwidth and cut-off frequency of an antenna are 0.8 MHz and 44 MHz respectively. Calculate the Q-factor of theantenna.
   (a) 55
   (b) 50
   (c) 75
   (d) 85
   (e) 65

4. Find v(0^+)$and(dv/dt)\_(0)+ifV(s) =$ [(-36s^2 - 24 s + 2) / (12 s^3 + 17 s^2 + 6 s)]
   (a) v(0^+) $= -4, (dv/dt)\_(0)+ = 2.25$
   (a) v(0^+) $= -2, (dv/dt)\_(0)+ = 3.25$
   (a) v(0^+) $= -3, (dv/dt)\_(0)+ = 4.25$
   (a) v(0^+) $= 0, (dv/dt)\_(0)+ = 1.5$
   (a) v(0^+) $= -3, (dv/dt)\_(0)+ = 1.25$

5. Evaluate f(t) if F(s) = [(3 s^2 + 17 s + 47) / {(s + 2)(s^2 + 4 s + 29)}]

(a) f(t) = [e$^-2t + 2.24e^-2tsin(5t - 26.6°)]u(t)$

(a) f(t) = [e$^-2t + 2.24e^-2tcos(5t - 26.6°)]u(t)$

(a) f(t) = [e$^-2t - 2.24e^-2tcos(5t - 26.6°)]u(t)$

(a) f(t) = [e$^-2t + 2.24e^-2tcos(5t + 26.6°)]u(t)$

True / False (10 marks)

Answer True or False

6. True or False: The inverse Laplace transform of V_ab(s) results in v_ab(t) = 16 e$^-3t + 34e^-2t + (33/2)e^-4t$.

6. True or False: The delta response h[n] of the given equation is h[n] = 2 (1/3)^n u[n] - 3 (1/3)^n u[n].

7. True or False: The steady-state solution of the given equation is y_ss[n] = $\sqrt{}$(3/7)cos[(/6)n + tan$^-1$(1/(2$\sqrt{3}$))].

7. True or False: The maximum pressure exerted on an aircraft flying at 200 mph at sea level is 15.42 psia.

8. True or False: When 100 J of work is done on a system of 1 mole of an ideal gas at constant temperature, the heat transferred is -100 J.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A cast-iron pulley rotates at 250 rpm and delivers 50 h.p. Considering the density of cast-iron is 0.05 lb/in^3 and the tensile stress is 150 lbs/in^2, derive the pulley diameter and discuss the implications of these calculations in engineering design.

12. Discuss the probability of obtaining a sum of 6 or 7 when tossing two dice. Analyze the possible outcomes and explain the calculation method used to arrive at your answer.

End of Paper